QUANT DEV / C++ / SYSTEMS

EXCHANGE LAB

06

PROPOSED RESEARCH / NOT YET VERIFIED

THE ENGINEERING QUESTION

What optimisations improve event processing latency without breaking matching determinism or accounting invariants?

PROBLEM DEFINITION AND RESEARCH CONTRIBUTION

A deterministic C++ limit order book matching engine used to examine the tradeoff between price time correctness, memory layout and tail latency.

- A basic order book is not enough; the value lies in rigorous testing and defensible benchmarking.
- You need a simple reference model, adversarial order flow and careful measurement discipline.
- This is one of the strongest zero cost portfolio pieces for your laptop.

ARCHITECTURE AND IMPLEMENTATION SPECIFICATION

1 STAGE **Order generator** → 2 STAGE **Risk + sequence** → 3 STAGE **Matching engine** → 4 STAGE **Journal + replay**

- C++20 single threaded reference implementation with integer tick prices, stable order IDs and deterministic event sequencing.
- Risk checks, journal/replay, cancellation, amendment, partial fills and property based fuzzing.
- Benchmark harness with warm up, repeat trials, compiler documentation and clear explanation of environmental controls.

ENGINEERING DECISIONS TO MAKE EXPLICIT

The design review must record data contracts, state transitions, failure handling and why the selected baseline is defensible. Every subsystem should have an interface specification and a reproducible test route.

DESIGN BOUNDARIES AND ASSUMPTIONS

⚠ Synthetic feed throughput does not equal production exchange readiness.

⚠ Microbenchmarks are easy to distort with poor methodology.

ⓘ Scope is proposed unless explicitly identified as documented existing work. All measured claims require reproducible evidence.

EXCHANGE LAB / PROTOCOL / ACCEPTANCE / REVIEW

EXPERIMENTS, VERIFICATION & PUBLICATION

06

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EXPERIMENTAL METHODOLOGY

01 Implement a simple reference engine and differential test optimised variants against it.

02 Generate adversarial workloads, varying price density, burstiness and cancellation behaviour.

03 Measure p50 / p99 / p99.9 latency, memory and replay determinism under stable benchmark conditions.

MEASUREMENT AND ACCEPTANCE EVIDENCE

- Matching invariants across adversarial sequences
- Documented p50 / p99 / p99.9 latency under fixed workloads
- Memory use and like for like baseline comparison

REQUIRED REPOSITORY ARTEFACTS

- Reference versus optimised correctness suite
- Reproducible benchmark harness
- Performance tradeoff report and demo results

TECHNICAL REVIEW QUESTIONS

Which assumptions could invalidate the result? How do the baselines behave under the same conditions? What is the strongest negative case? Which variables, seeds, workloads, software versions and machine properties must be retained for independent reproduction?

VERIFICATION MAPPING

For each acceptance claim, record the test identifier, input configuration, observed result, pass/fail definition and an independent rerun procedure. Failed or inconclusive cases remain in the report rather than being silently excluded.

PUBLICATION DISCIPLINE

Release original source, configuration files, test harnesses, raw measurements and scripts where rights permit. Distinguish simulation outcomes from physical validation and completed achievements from planned evaluation. Results must not be inferred from targets.